# Analysis Notes

Kota Uezono

June 6, 2026

**Theorem 0.1** (Hölder's inequality)**.** *Let* $p, q \in [1, \infty]$ *satisfy* $\frac{1}{p} + \frac{1}{q} = 1$ *and let* $f$ *and* $g$ *be measurable functions. Then*

$$\|fg\|_1 \le \|f\|_p \|g\|_q$$

*Proof.* The case $p = \infty$ is trivial. Assume $p, q < \infty$. We first prove the following auxiliary inequality: If $s, t > 0$, then

$$st \le \frac{s^p}{p} + \frac{t^q}{q}.$$

Let $\sigma, \tau \in \mathbb{R}$, and set $s = e^\sigma, t = e^\tau$. Since $e^x$ is convex and

$$\frac{1}{p} + \frac{1}{q} = 1$$

we obtain

$$e^{\sigma+\tau} \le \frac{e^{p\sigma}}{p} + \frac{e^{q\tau}}{q}.$$

Hence

$$st \le \frac{s^p}{p} + \frac{t^q}{q}.$$

Using this inequality for $(s, t) = (|f|, |g|)$, we obtain

$$|fg| \le \frac{|f|^p}{p} + \frac{|g|^q}{q}.$$

Integrating both sides, we obtain

$$\|fg\|_1 \le \frac{\|f\|_p^p}{p} + \frac{\|g\|_q^q}{q}. \tag{1}$$

Setting $C = \|f\|_p \|g\|_q, \tilde{f} = \frac{f}{\|f\|_p}, \tilde{g} = \frac{g}{\|g\|_q}$, we obtain

$$\|\tilde{f}\|_p^p = \|\tilde{g}\|_q^q = 1$$

The case $C = \infty$ is trivial. So, let $C < \infty$. Then,

$$C^{-1}\|fg\|_1 = \|\tilde{f}\tilde{g}\|_1 \overset{(1)}{\leq} \frac{\|\tilde{f}\|_p^p}{p} + \frac{\|\tilde{g}\|_q^q}{q} = \frac{1}{p} + \frac{1}{q} = 1.$$

Multiplying both sides by $C$, we obtain

$$\|fg\|_1 \leq \|f\|_p\|g\|_q.$$

$\square$